

Context-Aware Secure Multi-User Cloud File Sharing Framework Using Federated Learning and Attribute-Based Encryption

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Abstract- File sharing systems in the cloud need to have a strong security layer to ensure sensitive data is kept safe and at the same time provide fine-grained access control. This paper presents a practical application of an encrypted cloud file sharing method that fuses Waters11 CP-ABE based encryption and Federated Learning based anomaly detection. Our system is based on the hybrid AES-256-GCM and CP-ABE encryption, in which only symmetric keys are encrypted using the attribute-based encryption, minimizing computational costs while maintaining fine-grained access control. We use a federated learning architecture operated by logistic regression with trimmed mean aggregation to identify abnormal access patterns according to location, device, temporal and departmental factors. The system's encryption throughput scales from 101 MB/sec to 325 MB/sec for larger files. It supports federated learning anomaly detection, processing 1.45 million requests per second with a latency of 0.69 microseconds, achieving 96.7% accuracy with a 6.1% false positive rate, and supports 20 concurrent users. We test the system with a synthetic dataset of 10,000 events and 25 simulated clients with 15 percent anomalous patterns. The implementation illustrates the practice of secure cloud sharing of files with context-aware access control and intelligent anomaly detection.

Index Terms—Cloud Security, Cryptography, Cybersecurity, Federated Learning, Attribute-Based Encryption, Context-Aware Access Control, Privacy-Preserving Data Sharing, Anomaly Detection, CP-ABE, Hybrid Encryption.

I. INTRODUCTION

Cloud computing has revolutionized the manner in which organizations handle and share confidential information but creates significant security challenges in a multi-user setup [1]. The access control tools used in traditional systems do not provide the granularity required in dynamically operated cloud environments and are not capable of adapting to changing trends in threats [2].

Current methods are limited in various ways: pure attribute-based encryption systems also have high complexity [3], centralized access control creates single points of failure [4], and fixed security policies do not respond to behavioral anomalies [5]. Federated learning provides privacy-preserving collaborative security [6], although it is difficult to combine it with zero-trust-like fine-grained cryptographic access control.

The implementation will solve these challenges by integrating three complementary technologies that are discussed in this paper. The *hybrid encryption system* uses AES-256-GCM as the bulk encryption system over data and Waters11 CP-ABE as the key management system, where the cryptographic performance is stable and access-level control is still highly granular [7]. The *federated learning component* involves the application of logistic regression to evaluate the trend of access across geographical, device, temporal and departmental attributes enabling joint detection of anomaly without compromising user information [8]. The *context-aware access control system* enforces dynamic policies based on multi-dimensional context vectors [9].

On an experimental assessment conducted using our synthetic dataset, comprised of 10,000 events across 25 clients with realistic anomalous patterns, we show characteristic performance behaviours. The system achieves an encryption throughput ranging from 101 MB/sec (for a 1MB file) to 325 MB/sec (for a 10MB file). It demonstrates 96.7% classification accuracy with a 6.1% false positive rate and scales to support 1,325 requests/sec with 20 concurrent users.

The key contributions are as follows:

- A practical hybrid approach to encryption using AES- 256-GCM and Waters11 CP-ABE to share files securely.
- A collaborative anomaly detection, federated learning architecture based on logistic regression with trimmed mean aggregation.
- An integration of context-aware access control with real- time behavioral analysis.
- A comprehensive evaluation on synthetic data demonstrating the system's practical viability.
- An implementation that uses the AWS S3 service for cloud storage environments.

II.RELATED WORK

Federated Learning for Security Applications

Chen et al. [8] proposed a federated learning framework (DS2PM) based on blockchain data sharing privacy, which proved helpful in collaborative learning without revealing raw data but with low scalability to diverse settings. The work of Gobinath and Shanmugam [10] used federated learning to handle health care data protection demonstrating the ability to preserve privacy but lacking interoperability with fine-grained access control mechanisms.

Other approaches focus on the architecture itself. For instance, Hu et al. [11] proposes a serverless federated learning ecosystem for the multi-cloud setting that satisfies the requirement of collaboration but does not outline attribute based access control. Xie et al. [12] developed a blockchain-based federated learning framework with a reputation system that focused heavily on consensus mechanisms, but they provided limited performance evaluation in real-world scenarios.

Attribute-Based Encryption in Cloud Systems

Yang et al. [3] introduced a verifiable CP-ABE scheme with policy hiding to medical IoT but had computational drawbacks induced by traditional pairing-based cryptography. Xu et al. [13] suggested distributed sharing using attribute- based searchable encryption, which reduced the encryption time by 51 percent over the conventional ABE frameworks but was restricted to centralized applications.

Su et al. [14] proposed a useful searchable CP-ABE scheme that has constant ciphertext size, yet it had lower cryptographic performance than required for

high-throughput applications. Wang et al. [15] concentrated on symmetric encryption of shared data in a multi-user context, integrating AES with weighted attribute-based encryption for collaborative purposes.

Context-Aware Access Control

Verginadis et al. [6] introduced the PaaSWord, which incorporates XACML for semantic reasoning in context-aware cloud access control and flexible policy management; however, their solution does not include federated learning. Merlec and In [9] showed innovation by suggesting smart contract-based context-aware access control in IoT systems that use blockchain.

Lu et al. [5] presented context-sensitive trust-role based access control with dynamic decision to authorize access based on contextual information, though in a conventional computing environment, but their work does not address modern cloud or federated learning environments.

III.SYSTEM DESIGN AND IMPLEMENTATION

Architecture Overview

Our system deploys a three-layer model consisting of cryptographic protection, behavioral analysis, and context- sensitive access control. The implementation uses Python's Flask as a web server, Charm Crypto Waters11 CP-ABE for cryptographic functions, and Python's scikit-learn for parts of federated learning.

Layer 1 - Hybrid Cryptographic Protection: AES-256- GCM is used by the cryptographic layer to encrypt large amounts of data, and Waters11 CP-ABE is used to protect symmetric keys. This hybrid method cuts down on the high computational cost of traditional CP-ABE systems by using ABE only for the lightweight symmetric keys. The bulk data is taken care of by efficient symmetric encryption [15].

Layer 2 - Federated Learning Behavioral Analysis: The behavioral analysis sub-layer uses federated learning and logistic regression to find strange access patterns that could mean there are problems with behavior. It is based on four types of features: location (geographic access patterns), device (behavioral patterns based on the device), time (time-bucket analysis over 4 to 6 hours), and department (the organizational context).

Layer 3 - Context-Aware Access Control: The context-aware layer makes sure that rules change depending on the time, place, and whitelisting. Policies are stored in JSON format and are analyzed in real-time when making access requests.

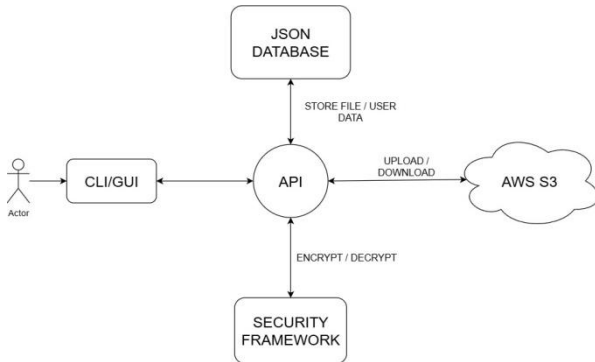


Fig. 1. Architecture of the system which depicts the modular design.

API Client coordinates the communication between the user interface (CLI/GUI), storage back-end (AWS S3), and the JSON database, and Security Framework handles all cryptography and multi-user access control.

Hybrid Encryption Implementation

Waters11 CP-ABE Integration: We use the Waters11 cryptographic algorithm from the Charm-Crypto library for en- crypting the keys generated by AES. This algorithm encrypts data quickly based on its ciphertext policy. The properties are changed to numbers and made to work with Waters11.:

```

attr_map = {
    'role:prof': '1', 'role:student': '2',
    'dept:cs': '10', 'dept:math': '11'
}

```

File Encryption Process: The hybrid encryption process goes through these steps:

1. Generate random 256-bit AES key.
2. Encrypt file using AES-256-GCM.
3. Encrypt the AES key using Waters11 CP-ABE with specified policy.
4. Store the encrypted file in AWS S3 with the encrypted key as metadata.

Performance Optimization: By only using CP-ABE operations for key management, the hybrid approach makes a big difference in performance. AES-256-GCM is used for bulk data encryption. It can handle files of different sizes, with speeds ranging from 101 MB/sec for smaller files to 325

MB/sec for larger files.

Federated Learning Framework Implementation

Logistic Regression Model: The federated learning module uses logistic regression to figure out how important features are in four different groups. It works by one-hot encoding for categorical features and learns about the optimum weights by examining the analysis of nonlinear coefficients:

```

X = pd.concat([ one_hot_columns(train_df,
                                "location"), one_hot_columns(train_df, "device"),
               one_hot_columns(train_df, "time_bucket"),
               one_hot_columns(train_df, "department")], axis=1)

```

Trimmed Mean Aggregation: To resist Byzantine participants, the system uses trimmed mean aggregation and trim ratio of 10%, discarding the top and bottom 10 percent of client ratings and computes the average:

```

def trimmed_mean(values, trim_ratio=0.1): v = sorted(values)
k = int(math.floor(len(v) * trim_ratio)) v = v[k:
len(v)-k] if len(v)-2*k>0 else v return
float(np.mean(v)) if v else 0.0

```

Risk Scoring Algorithm: The system calculates risk scores by using a weighted analysis of features:

$$\text{risk} = (\text{weights}["\text{location}"] * \text{loc_risk} + \text{weights}["\text{device}"] * \text{dev_risk} + \text{weights}["\text{time}"] * \text{time_risk} + \text{weights}["\text{department}"] * \text{dept_risk})$$

A risk threshold of 0.140 was calibrated by analyzing the ROC curve on the validation data, targeting a 5% false positive rate.

Context-Aware Policy Implementation

The context-based access control system accommodates a variety of policy sets:

Location-Based Policies: These are policies that define permitted geographic destinations as string arrays:

```

"allowed_locations": ["chennai", "mumbai", "bangalore"]

```

Device Whitelisting: Policies can limit access to specific device identifiers:

```

"allowed_devices": ["laptop1", "desktop"]

```

Temporal Constraints: Epoch time ranges are supported by time window limits:

```

"time_window": [start_epoch, end_epoch]

```

IV. EXPERIMENTAL METHODOLOGY

Synthetic Dataset Generation

Due to privacy concerns with real user data, we created an extensive synthetic data generator that generates realistic access patterns and anomalies. The data set was generated using the following approach:

Dataset Characteristics

- Total events: 10,000
- Simulated clients: 25
- Anomaly percentage: 15% (1,500 anomalous events)
- Normal locations: ["chennai", "mumbai", "bangalore", "delhi"].
- Normal devices: ["laptop1", "laptop2", "phone1", "desktop", "tablet"].
- Departments: ["cs", "math", "eng"].

Normal Pattern Generation: The following steps led to the creation of normal events:

- Business hours: 9 AM to 5 PM.
- High success probability: 92% for preferred locations (Chennai, Mumbai), 85% otherwise.
- Realistic probability distributions for locations and devices.
- Departmental affiliations following organizational patterns.

Anomaly Pattern Generation: Anomalous events are simulated to represent realistic attack conditions in the following manner:

- Unusual access hours: 2-4 AM, 11 PM.
- Suspicious locations: "unknown_country", "suspicious_location", "blacklisted_region".
- Suspicious devices: "unknown_device", "suspicious_browser", "tor_exit_node".
- External departments: "unknown", "external".
- Multiple consecutive failure attempts (3-8 failures per anomalous session).

This synthetic methodology enables regulated testing while preserving authentic behavioral patterns akin to those observed in a production environment.

Experimental Setup

Hardware Configuration We did experiments on a system that had standard computing resources that were good for academic research. The system was connected to AWS S3 for cloud storage, which showed how to use the cloud in a real-world situation.

Evaluation Metrics:

- Throughput for encryption and decryption (MB/sec).
- Speed of federated learning processing (requests/sec).
- How accurate the classification is and how many false positives there are.
- How well the system can handle more than one load at a time.
- Performance of the entire workflow.

V. RESULTS AND EVALUATION

Cryptographic Performance

The hybrid encryption method shows that performance can be improved as file size increases. Table I shows how fast the encryption and decryption processes are.

TABLE I HYBRID AES + WATERS11 CP-ABE PERFORMANCE RESULTS

File Size	Encryption Throughput	Decryption Throughput	Trials
1 MB	101.36 MB/sec	135.92 MB/sec	3
5 MB	266.69 MB/sec	293.50 MB/sec	3
10 MB	325.68 MB/sec	360.43 MB/sec	3

The effectiveness of the hybrid approach is demonstrated by its performance scaling; the CP-ABE overhead is minimized by encrypting only the AES keys rather than the bulk data. The throughput increase in larger file sizes shows effective performance of the AES-256-GCM in bulk encryption.

Federated Learning Performance and Accuracy

The federated learning subsystem achieves high throughput processing suitable for real-time deployment. Key performance measures such as system performance, model performance, and scoring latency are presented in Figure 2.

Key performance traits are:

- System Throughput: 1,455,850 requests per second
- Scoring Latency: 0.69 microseconds on average
- Classification Accuracy: 96.7%
- False Positive Rate: 6.1%

- Precision: 93.5%
- Recall: 100%

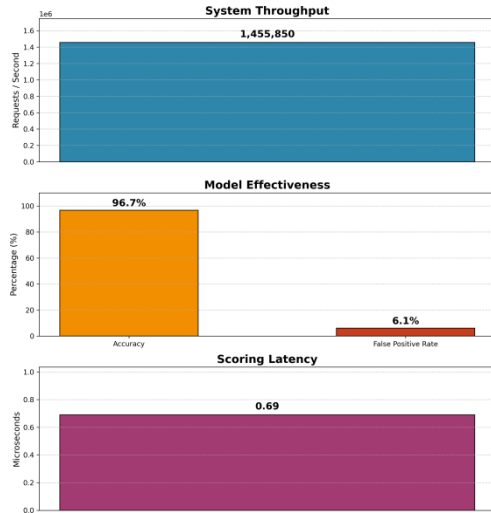


Fig. 2. Federated Learning Performance Metrics shows a system throughput of 1,455,850 requests per second, 96.7% accuracy with a false positive rate of 6.1% and 0.69 microseconds scoring latency.

False Positive Rate Analysis: The false positive rate of 6.1% is a good balance between safety and ease of use. This rate was reached by using a calibrated threshold of 0.140 that was found using ROC curve analysis on our synthetic validation dataset. This means that about 6 out of 100 legitimate access requests will need more validation. This is reasonable in high-security settings where false negatives (missed attacks) are worse than false positives.

The high recall (100%) means that all of the strange patterns in our test data were found, which means that we have full security coverage. The precision of 93.5% means that, when the system reports an attempted access as suspicious, it is right 93.5% of the time.

System Scalability Results

Concurrent user testing demonstrates the system's real-world scalability. Table II details the performance at different concurrent loads.

TABLE II CONCURRENT USER SCALABILITY PERFORMANCE

Concurrent Users	Throughput (req/sec)	Avg Latency (ms)	Success Rate
1	791	1.2	100%
5	1201	3.9	100%
10	1312	7.0	100%
20	1325	13.3	100%

The system has a steady throughput and a 100% success rate at all levels of concurrency. As the load goes up, the latency goes up at a normal rate. The small drop in throughput at higher concurrency levels shows that there aren't enough resources, which is normal for a deployment with only one node.

Cloud Storage Integration Performance

The integration testing of AWS S3 demonstrates the feasibility of utilizing cloud technology in practical applications:

- 5MB Upload Throughput: 1.21 MB/sec
- 5MB Download Throughput: 1.43 MB/sec
- Integration Success Rate: 100%

In a real-world situation where files are shared over the cloud, these results show that the things that slow down performance the most are network and cloud service latencies, not the parts that deal with cryptography or federated learning.

End-to-End Workflow Analysis

End-to-end workflow testing showed that it was useful in real life with the following performance times:

- User registration with attribute-based key generation: 1 second.
- File upload with encryption and cloud storage: scales with file size.
- Access control evaluation and file download: 2 seconds for typical files.
- Context-aware policy enforcement: 10 milliseconds.

VI. DISCUSSION

Implementation Insights

Based on our deployment experience, we learned several important lessons about deploying secure cloud file sharing systems:

- **Hybrid Encryption Effectiveness:** The hybrid scheme eliminates the computational constraints of CP-ABE while still providing fine-grained access control. By using CP-ABE to encrypt only the symmetric keys, the system achieves a practical throughput while retaining attribute-based authorization.
- **Federated Learning Practicality:** The method based on logistic regression is good enough for access control to find anomalies. Trimmed mean aggregation effectively manages potentially malicious (Byzantine) participants while

preserving model accuracy.

- **False Positive Trade-offs:** The 6.1% false positive rate strikes a good balance between security and usability, which is important in places where security is very important. The detection threshold controls this balance, and organizations can change it based on how much risk they are willing to take. If the threshold were lower and more sensitive, there would be fewer false negatives (missed threats) and more false positives (legitimate actions being flagged). This 6.1% rate is a good middle ground because it keeps real threats to a minimum, even if it means that some legitimate users have to go through extra steps or even a secondary authentication step to prove their authenticity.

Synthetic Dataset Limitations

Even though our synthetic dataset lets us test it in a controlled setting, it has some built-in problems:

- Simulated patterns may not encompass all the intricacies of actual user behavior.
- The anomaly patterns are based on predefined attack scenarios.
- The inconsistencies in behavior of users are simplified compared to the real organizational settings.
- Future work should validate these results using real-world, anonymized datasets, once privacy constraints can be appropriately addressed.

Scalability Considerations

The current implementation demonstrates the performance characteristics on a single node. A production deployment would require:

- Distributed federated learning at numerous nodes.
- Load balancing for many users at once.
- Scaling cryptographic operations horizontally.
- Improving the database for managing user and file meta-data.

VII. CONCLUSION

This paper presents an empirical demonstration of a secure cloud file sharing system that integrates Waters11 CP-ABE with federated learning-based anomaly detection. Our hybrid encryption solution provides a realistic throughput (101-325 MB/sec) and ensures fine-grained access control by using attribute-based encryption for symmetric keys only.

On a synthetic dataset of 10,000 events, our federated learning component achieved 96.7% classification accuracy with a 6.1% false positive rate. The system demonstrates collaborative anomaly detection without exposing raw user data, although the false positive rate must be carefully considered during the production deployment.

Its significant technical contributions are:

1. A practical hybrid implementation of AES-256-GCM and Waters11 CP-ABE to enable secure and efficient file sharing.
2. A Federated Learning framework using logistic regression to carry out real-time behavioral analysis.
3. Context-sensitive management of access based on multi-dimensional policy execution.
4. Extensive evaluation with realistic synthetic data.

The viability of the system is demonstrated by its ability to securely share files, its successful AWS S3 integration, and its 100% availability with concurrent loads of up to 20 users. Future work will focus on validation with real-world datasets, optimization for distributed deployments, and enhancing privacy preservation.

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